

```
In [24]: # Import required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.cluster import KMeans

# Load Iris dataset
iris = load_iris()

# Create DataFrame
df = pd.DataFrame(iris.data, columns=iris.feature_names)

print("First 5 Rows of Dataset:")
print(df.head())

# Select only two features for visualization
X = df.iloc[:, [0, 1]].values

# Create K-Means model
kmeans = KMeans(n_clusters=3, random_state=42)

# Train the model
kmeans.fit(X)

# Predict cluster labels
y_kmeans = kmeans.predict(X)

# Print cluster centers
print("\nCluster Centers:")
print(kmeans.cluster_centers_)

# Plot clusters
plt.scatter(X[:, 0], X[:, 1], c=y_kmeans, cmap='viridis')

# Plot centroids
plt.scatter(
    kmeans.cluster_centers_[:, 0],
    kmeans.cluster_centers_[:, 1],
    s=200,
    c='red',
    marker='X',
    label='Centroids'
)

plt.xlabel('Sepal Length')
plt.ylabel('Sepal Width')
plt.title('K-Means Clustering on Iris Dataset')
plt.legend()

plt.show()
```

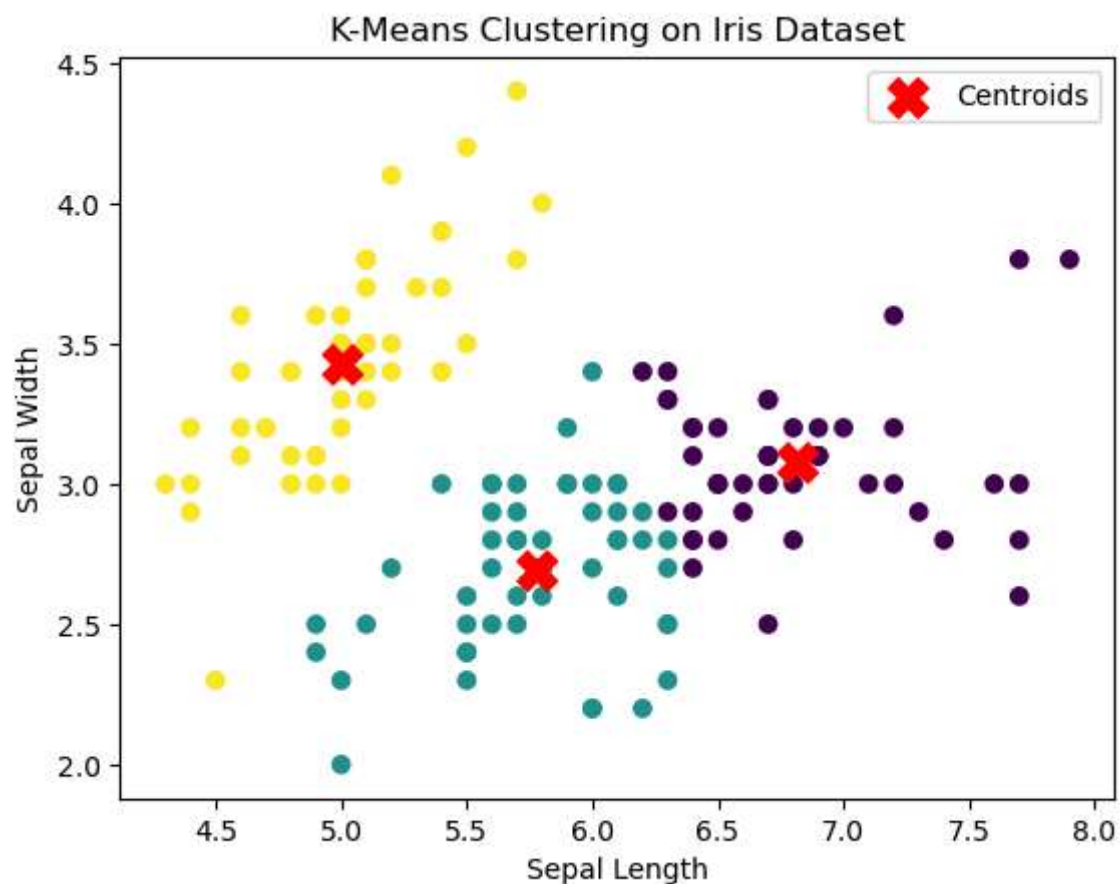
First 5 Rows of Dataset:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

Cluster Centers:

```
[[6.81276596 3.07446809]
 [5.77358491 2.69245283]
 [5.006      3.428      ]]
```

C:\Users\MGM\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1429: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
warnings.warn(



In []: